ABHISHEK PASHAPU

B.Tech Final Year | Electronics & Communication Engineering

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PROFESSIONAL SUMMARY

Final-year Electronics & Communication Engineering student with hands-on expertise in IoT, Embedded Systems, and AI-driven edge computing. Experienced in building production-grade systems including a live Raspberry Pi-based smart router with captive portal, web dashboard, and multi-device management for digital marketing deployments. Currently leading a real-time patient monitoring system using optimized CNN models on embedded hardware. Proficient in Linux, OpenWrt firmware, network security, and cloud-edge integration. Seeking full-time or internship roles in IoT Engineering, Embedded Systems, Firmware Development, or Network Security.

TECHNICAL SKILLS

Embedded & IoT	Raspberry Pi, OpenWrt, Arduino Uno, RTOS Fundamentals, Firmware Design, Edge Computing
Networking	Wi-Fi Access Systems, Captive Portal (OpenNDS), Network Security, Bandwidth Control, TCP/IP
AI / ML	CNN Model Optimization, Quantization, TensorFlow Lite, Edge AI, Real-Time Inference
Programming	Python, C/C++, Bash/Shell Scripting, JavaScript
Backend & Cloud	REST APIs, Backend Connectivity, Cloud-Edge Integration, Database Integration
Tools & OS	Linux (Ubuntu/Debian), Git, Web Dashboard Development, Technical Documentation
Cybersecurity	Kali Linux, Wireshark, Metasploit, Ethical Hacking, Penetration Testing, Vulnerability Assessment, Network Security, SQL Injection, Session Hijacking, Phishing Analysis, Wi-Fi Hacking, Web Penetration Testing, IoT Hacking, Proxy & VPN, Cloud Security, Footprinting & Reconnaissance

EXPERIENCE

IoT Engineer Intern | Devlats Pvt. Ltd. | Nizamabad, Telangana *Jun 2025 – Present*

- Developed and deployed IoT solutions for smart home applications, managing end-to-end hardware and software integration.
- Built and maintained a live smart router system using Raspberry Pi and OpenWrt to manage multiple user devices and enforce network access control.
- Implemented captive portal authentication (OpenNDS) with secure user management, bandwidth control, and digital marketing integration for real-world deployments.
- Designed and deployed a web-based dashboard for real-time monitoring and management of connected devices and network traffic.
- Collaborated with engineering teams to test device connectivity, performance, and functionality across IoT deployments.
- Documented project workflows, technical specifications, and system architecture to support team collaboration and future maintenance.

Cybersecurity Trainee | GradGuru | Remote *Jan 2025 – Mar 2025*

- Completed intensive hands-on cybersecurity training covering ethical hacking, penetration testing, network security, vulnerability assessment, session hijacking, web penetration testing, Wi-Fi hacking, cloud security, IoT hacking, proxy/VPN configuration, and SQL injection.
- Gained practical experience with industry-standard tools including Kali Linux, Wireshark, and Metasploit for real-world attack simulation and network analysis.
- Performed footprinting and reconnaissance using OSINT tools and open-source websites to gather target information as part of structured ethical hacking exercises.

- Conducted phishing attack simulations and studied social engineering techniques to understand and defend against real-world threat vectors.
- Built a personal home lab using a secondary system and legal practice platforms to independently test and apply all learned concepts including SQL injection, session hijacking, and Wi-Fi security auditing.

PROJECTS

IoT-Based Smart Router – Network Security & Access Control | | *Jun 2025 – Present*

Role: IoT Developer & Hardware Lead

- Designed and deployed a production-level smart router using Raspberry Pi and OpenWrt firmware capable of managing multiple concurrent user devices.
- Integrated OpenNDS captive portal for secure user authentication, session management, and digital marketing display on network login.
- Built a full-stack web dashboard for real-time network monitoring, device visibility, bandwidth control, and admin management.
- Implemented secure Wi-Fi access policies, firewall rules, and automated scripts for robust network security in live deployment environments.

Hybrid Edge-Cloud Architecture for AI on Embedded Systems | | *May 2025 – Present*

Role: Team Lead – Hardware & Software Integration

- Leading a team to design a hybrid architecture that offloads computational constraints from embedded devices to cloud infrastructure for AI model execution.
- Overseeing hardware-software integration to enable real-time patient monitoring using optimized and quantized CNN models on resource-constrained embedded devices.
- Developing efficient edge inference pipeline using TensorFlow Lite to minimize latency while maintaining model accuracy for offline AI processing.

Smart Car Safety System using Arduino Uno | | *Jan 2025 – Apr 2025*

Role: Lead Developer

- Developed an embedded safety system for vehicles using Arduino Uno incorporating sensor-based hazard detection and automated alert mechanisms.
- Implemented real-time sensor data acquisition, processing logic, and actuator control for collision avoidance and driver safety alerts.

EDUCATION

B.Tech – Electronics & Communication Engineering *Nov 2022 – Present*
Kshatriya College of Engineering, Armoor

Intermediate (M.P.C) *Jul 2020 – Mar 2022*
Kshatriya Junior College, Armoor

Secondary School Certificate (S.S.C) *Mar 2019 – Mar 2020*
Nalanda High School, Armoor

ADDITIONAL INFORMATION

Languages: English (Professional), Telugu (Native), Hindi (Conversational)

Core Strengths: Problem-Solving, Team Collaboration, Self-Directed Learning, Technical Documentation, Project Management

Availability: Immediate – Open to Internships & Full-Time Roles